From Notation to Gesture: Virtual Conductor Gesture Generation in VR via Structured Score Semantics

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Figure 1: Our method generates the gesture movements of a symphony conductor by learning hidden features such as the tempo and time signature of a musical score, which are then applied to VR scenes. First, users can select a piece they wish to enjoy or practice from the playlist. Next, they can position themselves freely according to their instruments. Simultaneously, the musical score is displayed within their field of view for them to follow and observe. Finally, the conductor's movements are shown, allowing the user to replicate the gestures to learn within the scene.

Abstract

Conductor avatar plays a dual role in immersive Virtual Reality (VR) interactive systems by interpreting musical scores and guiding orchestral performance. Rule-based score-driven methods ensure precise synchronization with predefined conducting templates or videos, but are constrained by pre-authored data. Audio-driven frameworks offer greater adaptability through real-time gesture generation but often fail to capture the symbolic semantics of musical scores. To overcome these limitations, we propose a novel score-driven gesture generation framework that translates symbolic musical representations into plausible conducting gestures. Our approach adopts a two-stage architecture, combining a comparative learning stage for pre-training a score encoder with a generative learning stage for gesture synthesis. The score encoder explicitly models musical features such as tempo, chord, intensity, and cycle semantics, directly informing gesture generation. To support this research, we introduce Multimodal Symphonic Conducting Dataset (MSCD), the first synchronized dataset comprising conducting gestures, performance audio, and editable symbolic scores, effectively bridging the gap between musical semantics and gesture synthesis. Qualitative and quantitative analyses are provided to demonstrate the effectiveness of our approach, while a user study is designed to identify the strengths and limitations of the current work.

1 Introduction

The rapid advancement of digital media technology has profoundly reshaped traditional artistic expression, with immersive VR environments enabling novel forms of interactive experience across artistic domains [9, 31, 32, 50]. Symphonic, as a cornerstone of classical music, is also embracing this transformation through digital reinterpretation within immersive contexts [10, 29]. Within this evolving landscape, conductor avatars play a pivotal role as mediators between symbolic musical scores and expressive interpretation [36, 58].

In an orchestra, performers observe the conductor using their peripheral vision while concentrating on their musical scores. This reliance makes screen-based interfaces, like tablets, ineffective for continuous monitoring. Furthermore, the subtle spatial relationships between each musician and the conductor are vital for maintain-

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ing collective synchrony. Therefore, achieving immersive spatial fidelity is essential, and VR offers this capability more effectively than traditional media. Consequently, developing VR-based conductor systems that can replicate the dynamics between conductors and performers, utilizing immersive spatial perception, presents a significant challenge in the digitization of symphonic music.

Current efforts to digitize conducting fall into two primary categories: rule-based methods [6, 41] and audio-driven frameworks [34, 60]. Rule-based approaches rely on predefined gesture templates or videos that are synchronized with symbolic musical scores. For example, VR applications such as Maestro [21] and OSR Virtual Hall have demonstrated the feasibility of immersive conducting interactions by aligning preauthored gestures with preloaded scores. These systems validate the potential of score-guided conducting in immersive environments but lack flexibility and adaptability. In contrast, recent research has increasingly adopted audio-driven frameworks [7, 60], which generate conductor gestures from audio features such as Mel spectrograms. These approaches enable real-time synchronization and adaptability to performance variations, demonstrating notable improvements in responsiveness. Despite these advances, several key challenges remain unresolved. Rule-based methods rely on rigid, non-adaptive gesture sequences or videos that fail to capture the nuanced relationship between score semantics and conducting gestures. Meanwhile, audio-driven frameworks introduce unavoidable latency and offer limited control over gestures through symbolic manipulation. Furthermore, although these approaches effectively reflect surface-level rhythmic and dynamic patterns, they fall short in interpreting higher-order musical semantics embedded in scores, such as harmonic tension, phrase structure, and expressive contour.

To address these limitations, we present a **novel score-driven framework** for conducting gesture generation that grounds physical movement in *symbolic musical semantics*, and the application showcase is shown in Fig. 1. Rather than relying on audio signals or fixed templates, our approach directly learns from the structural and expressive elements embedded in the score itself. The framework is built on a two-stage architecture adapted from Diffusion-Conductor [60], designed to capture both the semantic richness and temporal structure of symphonic music. In the contrastive learning stage, we introduce a hierarchical score encoder to disentangle and model key musical dimensions, including tempo, intensity, chord progression, and cyclical structure. This modular design enables fine-grained representation learning and establishes a meaningful alignment between symbolic inputs and gesture dynamics. In the generative modeling stage, we incorporate a Denoising Diffusion Implicit Model (DDIM), a Cross-Modality Linear Transformer, and a random masking strategy. Together, these components synthesize temporally coherent and contextually appropriate gestures conditioned on symbolic score embeddings. This architecture can generate expressive and musically grounded conducting motions that are adaptable to a variety of orchestral contexts.

Moreover, we introduce the **MSCD dataset**, a new benchmark designed to support score-driven conductor gesture generation. We curated concert videos featuring conductor-focused views and retrieved their corresponding musical scores. These scores were converted into MuseScore format and manually aligned with the video data to ensure frame-level temporal synchronization. Using OpenPose, we extracted 2D keypoints to represent conducting gestures, resulting in a richly annotated multimodal dataset. To the best of our knowledge, MSCD is the first publicly available dataset that jointly provides conductor gestures, performance videos, and editable symbolic scores. It fills a critical gap by enabling supervised learning of gesture synthesis grounded in symbolic musical semantics.

Finally, we conduct quantitative and qualitative studies to evaluate the quality, semantic consistency, and readability of our generated gestures. The results validate our system's ability to generate expressive and structurally coherent conducting gestures in response to symbolic score features. In summary, our work redefines gesture generation for conductor avatars by grounding it in score semantics rather than audio reactivity. This paradigm shift preserves the conductor's dual role as both architect and interpreter of musical expression. It also offers a valuable resource for broader research on multimodal alignment and symbolic-to-physical generation in music-centered applications.

The main contributions of this work are as follows:

- We present a novel score-driven gesture generation framework that generates conductor gestures based on symbolic musical representations, which enable structured, interpretable, and controllable modeling of musical intent.
- We introduce MSCD, the first synchronized dataset containing conducting gestures, performance audio, and editable symbolic scores, enabling future research on multimodal generation and interpretable embodied interaction.
- We conduct quantitative and qualitative studies to validate the aesthetic quality, semantic consistency, and practical readability of the gestures generated by the proposed framework.

2 Related Works

2.1 Symbolic Music Encoding

Mainstream symbolic music encoding methods include piano roll-based and MIDI-based approaches. MIDI-based methods have gradually replaced piano roll-based methods due to the fixed-length data representation and improved computational efficiency. MIDI-like encoding methods [22, 40] imitate the MIDI protocol and encode music pieces with note-on, note-off, time-shift, etc. Huang et al. [24] proposed REMI, which replaces note-off events with note duration. This framework also integrates structured semantic features such as bar, position, chord, and tempo, improving the density of symbolic music information. However, the long sequence lengths generated by this encoding framework posed significant computational challenges in processing complex, extended compositions such as symphonies. To address these limitations, Hsiao et al. [20] proposed the Compound Word (CP) method, which restructures REMI's discrete token sequences into compact representations based on compound temporal units, effectively reducing sequence redundancy. Building on the CP framework, Li et al. [49] incorporated an N-gram probabilistic model to capture temporal correlations between adjacent musical events, thereby enhancing the efficiency and accuracy of sequence modeling. At the same time, Zeng et al. [59] introduced the OctupleMIDI encoding system, achieving breakthroughs in multi-track music representation. This methodology encodes each note as an 8-tuple encompassing time signature, tempo, bar, position, instrument, pitch, duration, and velocity parameters, simultaneously compressing sequence dimensionality and preserving harmonic relationships across tracks. We follow the OctupleMIDI encoding framework to encode score features, enabling fine-grained modeling of score structures. The encoding method will be introduced in Sec. 3.3.

2.2 Virtual Conductor

Previous research has explored the topic from multiple perspectives. Baez et al. [4] proposed a system that emulates an ensemble conductor for string quartets; P Bos et al. [6] present a virtual embodied agent that can conduct musicians in a live performance. Funk et al. [14] discussed whether a virtual conductor can generate an emotionally associated interpretation just like a real human did through research and investigation. Ter Maat et al. [47] designed and implemented a Virtual Conductor that acts like a human conductor in its interaction with a realistic human orchestra. Chen et al. [7] present a work that can generate conductor gestures from any given music and a large-scale dataset with music and gesture. Zhao et al. [60]

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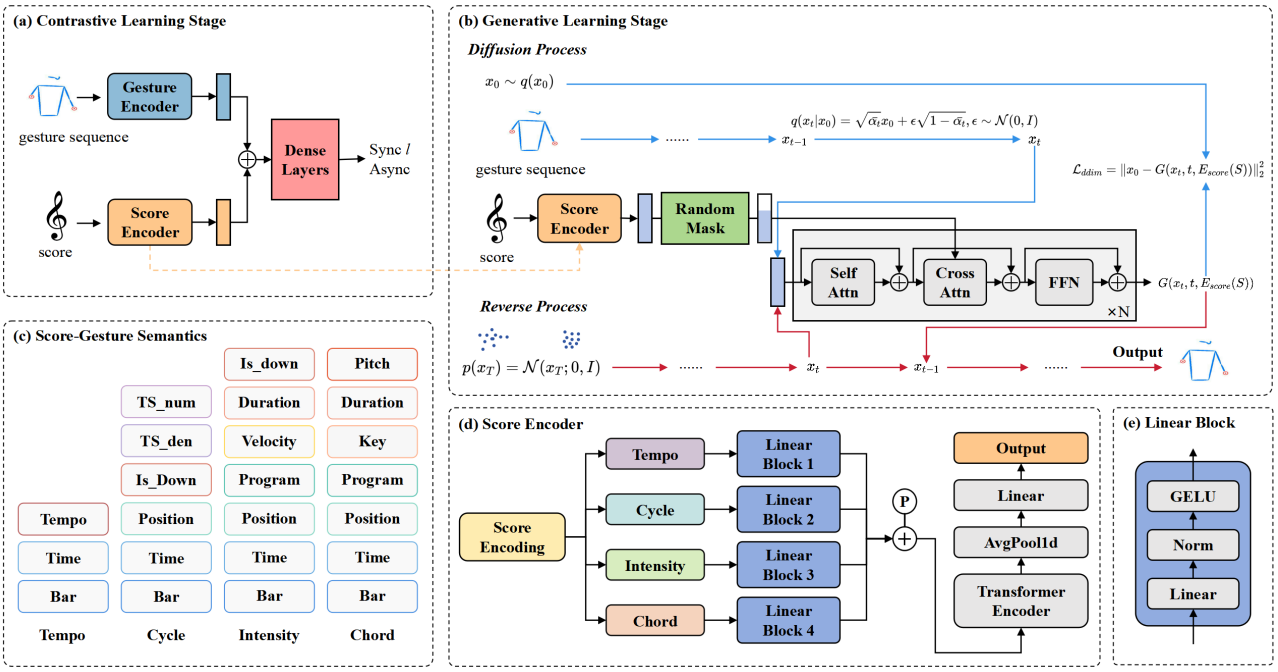


Figure 2: The overall architecture. We use the two-stage architecture to generate the conductor gestures: (a) the contrastive stage will pre-train the score encoder; (b) a DDIM framework with a Cross-Modality Linear Transformer and a random masking strategy to generate the gestures; (c) four dimensions semantics of scores corresponding to the gestures; (d) the score encoder; (e) the linear block in the score encoder.

followed the previous work and introduced the diffusion model, and Oh et al. [39] introduced transfer learning, which both led to great results. Previous work overlooked the structural semantics encoded in musical scores, which are crucial for governing real-world conducting practice. Our work addresses this gap by developing a score-driven framework for generating conductor gestures. Our approach enables the modification of score parameters to influence how conducting gestures are generated.

2.3 Gesture Generation

The field of gesture generation has evolved through three main paradigms: rule-based systems, statistical models, and deep learning techniques, each marking significant advancements in how gestures are synchronized with other signals. Early systems relied on manually defined rules to synchronize gestures with speech or text. Statistical models learned latent gesture distributions from annotated data. The current shift toward deep learning has enabled data-driven gesture synthesis at scale. Audio-driven approaches have been widely explored, with Hasegawa et al. [18] pioneering the use of bidirectional LSTMs for autoregressive gesture generation from speech audio. This work established a foundational approach for modeling temporal consistency between audio and gesture. Kucherenko et al. [26] subsequently introduced representation learning via autoencoders to disentangle temporal smoothing. Further advancements include normalizing flow methods [1], VAE-based approaches [3, 15, 16, 33, 46], hybrid systems [13, 17, 28, 61], GAN-based [12, 43, 44, 54] and diffusion-based frameworks [23, 52]. Text-driven approaches emerged to address the semantic limitations of audio-only methods, employing similar architectures. Notable works include Ishi et al. [25], Yoon et al. [57], and Bhattacharya et al. [5], which focused on enhancing gesture meaningfulness through textual semantics. Recognizing the limitations of single-modality inputs, recent efforts have focused on integrating multimodal signals, enhancing the richness and accuracy of gesture generation. Yoon et al. [56] proposed the first trimodal framework combining text, audio, and style inputs for gesture synthesis. The advent of transformer architectures [51] and BERT [11] further advanced semantic text encoding, inspiring works such as Ahuja et al. [2] and Kucherenko et al. [27]. Recently, diffusion models have gained prominence in gesture synthesis [8, 19, 37]. Our work builds on established gesture generation frameworks and evaluation metrics.

3 Methodology

In this section, we will first provide an overview of our work. Then, we will present the problem formulation. Next, the encoding method will be introduced. Finally, we will introduce the various stages and the training objectives.

3.1 Overview

Our work extends the two-stage architecture established by Zhao et al. [60], as illustrated in Fig. 2. During the contrastive learning stage, we utilize a gesture encoder $E_{gesture}$ and a score encoder E_{score} to learn latent representations of musical score information and align them with the corresponding gesture embeddings. In the subsequent generative learning stage, we leverage the pre-trained score encoder E_{score}, a Denoising Diffusion Implicit Model (DDIM), and a cross-modality linear transformer to construct the denoising process for synthesizing gesture sequences. During inference, Gaussian-distributed noise, along with a specified random seed and score features, will be input into the denoising process to generate score-driven conducting gesture sequences.

3.2 Task Definition

Our goal is to generate realistic and plausible conductor gestures $\hat{M} = \{M_1, M_2, ..., M_n\}$ which should be similar to the original gestures M, with the score $S = \{S_1, S_2, ..., S_t\}$ as input, where M_i represents one pose array at frame i, n is the length of gesture sequences, S_q represents one note and k is the length of score note sequences. For each $M_i \in \mathbb{R}_{J \times D}$, J is the number of joints, and D is the joint dimension. Score feature $E_{score}(S)$ will be extracted by Score Encoder E_{score}. A diffusion model G will be trained to generate a gesture

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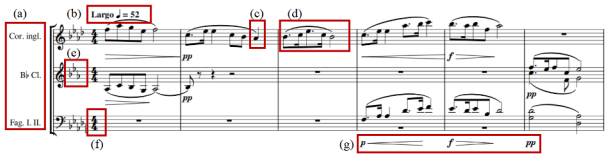


Figure 3: An excerpt from the second movement of Dvořák's Symphony No. 9 "From the New World," illustrating key elements in the musical score: (a) program; (b) tempo; (c) a note with duration and pitch information; (d) a complete measure; (e) key signature; (f) time signature; (g) expression markings affecting velocity.

sequence M with the corresponding $E_{score}(S)$. The task formulation can be written as:

$$\underset{G,E_{score}}{argmin} \|M - G(E_{score}(S))\| \tag{1}$$

3.3 Score Encoding Method

As illustrated in Fig. 3, our encoding framework primarily focuses on the processing of musical score annotations. Notably, we utilize a MusicXML-like notation format through the PDMX system proposed by Long et al. [35], which implements an efficient conversion pipeline that transforms MuseScore-compatible MSCZ files into a structured JSON format. The PDMX framework first converts MSCZ files into a MusicXML-like intermediate representation, preserving the symbolic semantics of musical content, such as pitch and tempo. Building on this foundation, our methodology expands feature extraction to eleven dimensions, consistent with the terminology used in MusicXML: *measure*, *time*, *program*, *position*, *pitch*, *velocity*, *duration*, *is_downbeat*, *tempo*, *key signatures*, and *time signatures*. Time signatures are stored as both a numerator and a denominator.

Time. In MusicXML-like score representations, temporal alignment is achieved through the coordinated use of *time* and *resolution* parameters. Here, *resolution* defines a quarter note in relative temporal units, typically set to 480. *Time* denotes the cumulative count of these units from the score's origin. This configuration establishes proportional duration relationships; for instance, an eighth note corresponds to 240 units ($0.5 \times resolution$). These values represent symbolic durations rather than absolute time. This relationship can be formulated as $time = N_q \times resolution$, where N_q denotes the number of quarter notes from the beginning of the score.

Measure and Position. The *measure* field corresponds to a bar in music, with position represented as an offset in quarter-note units. The range $[0, N)$ is based on the time signature, where N indicates the beats per measure.

Program. The *program* field indicates the instrument designation per the MusicXML specification, using integer codes from 0 to 127.

Pitch, Velocity, and Duration. These three parameters define the acoustic state of a note. *Pitch* follows the standard MIDI numbering system, ranging from 0 to 127, where 60 represents middle C. *Velocity* measures the dynamic intensity of a note's onset within a range of 2 to 128, with higher values indicating stronger attacks. *Duration* is calculated by multiplying the note type by the resolution, in alignment with how *time* is measured.

Key Signature. Tonal centers are represented as integers according to the MusicXML convention, where C major is assigned the value 0. Positive and negative integers indicate sharp and flat key signatures, respectively.

Tempo. Expressed as a floating-point value representing quarter-note duration in beats per minute (BPM). This continuous representation accommodates tempo fluctuations and supports alignment across performance variations.

Is_downbeat. A Boolean flag that identifies metrically strong beats, typically marking the start of each measure.

Time Signature. Divided into two integer parts: the numerator indicates beats per bar, while the denominator specifies the beat unit.

3.4 Contrastive Learning Stage

The contrastive learning stage comprises three components: a gesture encoder $E_{gesture}$, a score encoder E_{score}, and dense layers f_{CL}.

The gesture encoder adopts the Spatial-Temporal Graph Convolutional Network (ST-GCN) [55] to extract spatio-temporal features from the skeletal joint trajectories through graph-based convolutional operations across temporal and spatial dimensions, proven effective in prior pose estimation tasks.

The score encoder processes hierarchical musical score representations by explicitly separating inputs into four categories: tempo context, cycle context, intensity context, and chord context. These features are aligned by bar and time. Each feature subset undergoes an independent linear projection with layer normalization and GELU activation before concatenation. The combined features are then further encoded using positional embeddings and a transformer encoder that consists of 2-layers and 4-heads. Finally, dimensional uniformity is achieved through average pooling and linear projection.

The contrastive learning network takes concatenated embeddings from $E_{gesture}$ and E_{score}. It optimizes cross-entropy loss by aligning positive pairs (matched score-gesture pairs) and performing negative sampling from mismatched instances within the training batch. The process establishes a cross-modal correspondence between score semantics and conductor kinematics. The formulation is written as:

$$\begin{aligned} L_{bce} = \sum_{i,j=1}^{N} & (c_{ij} log_2(f_{CL}[E_{gesture}(M_i) \oplus E_{score}(S_j)]) \\ & + (1 - c_{ij}) log_2(1 - (f_{CL}[E_{gesture}(M_i) \oplus E_{score}(S_j)])) \end{aligned} \tag{2}$$

where M_i and S_j represent the i-th gesture data and j-th score data, $\oplus$ represents the concatenation operation. c_{ij} is defined by

$$c_{ij} = \begin{cases} 1 & i = j \\ 0 & otherwise \end{cases} \tag{3}$$

3.5 Generative Learning Stage

The Generative Learning Stage incorporates three key components: a diffusion-based generative model, a Cross-Modality Linear Transformer, and a random masking strategy.

The diffusion process is defined as a discrete-time Markov chain [19]; given an initial gesture sequence x_0, the noise-adding process is formulated as:

$$q(x_{1:T}|x_0) = \prod_{t-1}^{T} q(x_t|x_{t-1}) \tag{4}$$

where the single-step transition kernel follows. where β_t is a noise scheduling parameter for each step of the forward diffusion process, controlling the amount of noise added at each step, which is a pre-set hyperparameter:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \tag{5}$$

Through the reparameterization trick, define $\alpha_t = 1 - \beta_t$ to denote the proportion of the original signal retained at step t, and $\bar{\alpha}_t = \prod_{s=1}^{t} a_s$ to denote the cumulative product. The closed-form solution at any time step t is derived as:

$$q(x_t|x_0) = \sqrt{\bar{\alpha}_t}x_0 + \varepsilon\sqrt{1-\bar{\alpha}_t}, \varepsilon \sim \mathcal{N}(0, I) \tag{6}$$

Departing from conventional noise prediction approaches [19,38], the model directly regresses the original data by using the mean square error on this prediction. θ is the parameter of the neural

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Table 1: Comparison on the Conducting Motion Datasets

Dataset	Pose Source	Score	Tracks	Conductor	Time(min)
Sarasúa et al.	Motion Capture	—	10+	1+	120
Sarasúa and Guaus et al.	Motion Capture	Beats	1	**25**	250
Karipidou et al.	Motion Capture	—	1	1	36
Huang et al.	Motion Capture	Velocity, Tempo, Note	3	6	180
Lemouton et al.	IMU	—	1	2	56
Chen et al.	Object Detection	—	**100+**	10+	**6000**
Ours	Object Detection	**Whole Score**	39	11	348

network in the reverse denoising process, μ_θ is the mean and Σ_θ is the variance of the neural network prediction. The reverse process approximates the posterior distribution:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t,t), \Sigma_\theta(x_t,t)) \tag{7}$$

Furthermore, the random mask is used after the score encoder to train the diffusion model with both score-conditional and unconditional elements. In the meantime, the denoising model adopts the standard Transformer framework [51], utilizing a pre-trained score encoder to extract embeddings, with self-attention modules modeling temporal gesture dependencies and cross-attention mechanisms fusing score-gesture features. The final gesture sequence is generated through a feed-forward network $G(x_t, t, E_{score}(S))$.

Above all, the overall loss function of the generative learning stage can be written as:

$$\mathcal{L} = \lambda_{ddim}\mathcal{L}_{ddim} + \lambda_{perc}\mathcal{L}_{perc} + \lambda_{geo}\mathcal{L}_{geo} \tag{8}$$

where $\mathcal{L}_{ddim}$ represents the diffusion loss, $\mathcal{L}_{perc}$ represents the perceptual loss and $\mathcal{L}_{geo}$ represents the geometric loss. λ_{ddim}, λ_{perc}, and λ_{geo} are weights of each loss.

As mentioned above, we adopt direct gesture prediction [42, 48] instead of noise estimation [19] in the diffusion process, formulating the diffusion loss as:

$$\mathcal{L}_{ddim} = \|x_0 - G(x_t, t, E_{score}(S))\|_2^2 \tag{9}$$

where x_0 denotes ground-truth gesture sequences and $G(\cdot)$ represents the final denoising step.

A perceptual loss is used to minimize the distance between the extracted feature from the generated gesture and the ground-truth gesture.

$$\mathcal{L}_{perc} = |E_{gesture}(x_0) - E_{gesture}(\hat{x_0})| \tag{10}$$

where $E_{gesture}$ is the pre-trained gesture encoder in the contrastive learning stage, and $\hat{x_0} = G(x_t, t, E_{score}(S))$.

Additionally, geometric constraints ensure both physical plausibility and artistic coherence. The velocity loss aligns kinematic speeds with reference data while the elbow loss regulates the natural patterns of arm oscillation.

$$\mathcal{L}_{geo} = \lambda_{vel}\mathcal{L}_{vel} + \lambda_{elbow}\mathcal{L}_{elbow} \tag{11}$$

$$\mathcal{L}_{vel} = \frac{1}{N-1}\sum_{i=1}^{N-1}\left\|(x_0^{i+1} - x_0^i) - (\hat{x_0}^{i+1} - \hat{x_0}^i)\right\|_2^2 \tag{12}$$

$$\mathcal{L}_{vel} = -\frac{1}{N-1}\sum_{i=1}^{N-1}\left\|(\hat{x_0}_{elbow}^{i+1} - \hat{x_0}_{elbow}^{i})\right\|_2^2 \tag{13}$$

where λ_{vel} and λ_{elbow} are weights for each term.

4 Dataset

In this section, we will introduce MSCD, detailing its sources, preprocessing procedures, alignment strategies, and statistical analysis.

4.1 Data Collection and Annotation

The dataset primarily comprises videos sourced from publicly available platforms such as YouTube and BiliBili, ensuring diversity in conductors, repertoire, and conducting styles. To ensure data quality and consistency, we apply the following filtering criteria: videos with unstable viewpoints or shifting camera angles are excluded and low-resolution videos are removed due to unreliable keypoint extraction performance. Musical scores are primarily collected from the MuseScore website, emphasizing open-access materials. For cases where editable scores are unavailable, non-editable PDF versions are retained, which will be manually converted into MuseScore format.

During preprocessing, we crop the video to focus on the conductor's primary movements and standardize all videos to a resolution of 1920×1080 at 25 frames per second. For recordings that contain multiple symphonic movements within a single video, temporal segmentation is applied to isolate each movement into a separate clip, facilitating alignment. A key preprocessing step involves handling the preparatory gesture before the first beat, which varies across recordings. To ensure consistency, we define a temporal selection window that begins once the conductor completes the preparatory stance and ends at the onset of the first beat.

We use OpenPose to extract 2D upper-body keypoints, including the left/right eyes, ears, shoulders, elbows, wrists, hips, and nose. Each frame is represented as a tensor of shape [13, 2], corresponding to the coordinates of 13 keypoints. Then, the normalization of the data is carried out using the following method: First, the hip joint point is designated as the origin to calculate offsets, allowing all data to be translated accordingly. Next, pose coordinates are converted into vector representations, followed by a normalization process to remove the effects of bone length variations. Finally, the vectors are transformed back into pose coordinates using the inter-bone length measurements referenced from previous gesture generation studies.

Alignment of the video and score requires domain expertise and follows a three-step pipeline. First, the tempo markings in the score are manually adjusted in MuseScore to match the observed tempo of the performance. Second, any unperformed segments are manually identified and removed from the score. Finally, alignment anchors are established at 10-second intervals along the timeline to aid multimodal synchronization.

As of writing, we have collected 61 paired scores in total, the final dataset comprises 28.3 hours of conductor video recordings. Of these, 12.2 hours are unpaired due to missing score annotations, and 4.7 hours are accompanied only by PDF-format scores. The fully processed and aligned paired subset includes 11.4 hours of data, covering 37 musical movements performed by 11 conductors. The dataset will be published in https://sites.google.com/view/scoreconductor/.

4.2 Data Analysis

Table 1 compares our dataset with existing datasets related to conductor motion analysis. While our total duration is comparable to other public benchmarks, our dataset distinguishes itself through its multimodal richness—offering synchronized video, editable symbolic scores, and keypoint annotations. This diverse modality coverage provides a comprehensive foundation for research.

In our analysis, musical bars serve as the basic statistical unit. Figure 4(a) shows the distribution of instrument usage across the dataset. As expected, string instruments constitute the largest portion, with violins accounting for 18.4%, violas 8.3%, cellos 10.3%, and double basses 7.3%. Woodwinds, typically used for melodic and ornamental passages, include flutes (11.1%), oboes (6.8%), clarinets (6.7%), and bassoons (9.0%). Brass instruments, which often contribute to climactic or rhythmic sections and generally exhibit shorter durations, include horns (7.8%), trumpets (4.1%), trombones (2.8%), and tubas (0.8%). Percussion instruments emphasize rhythm and dynamics, accounting for a total of 4.1%. Other instruments,

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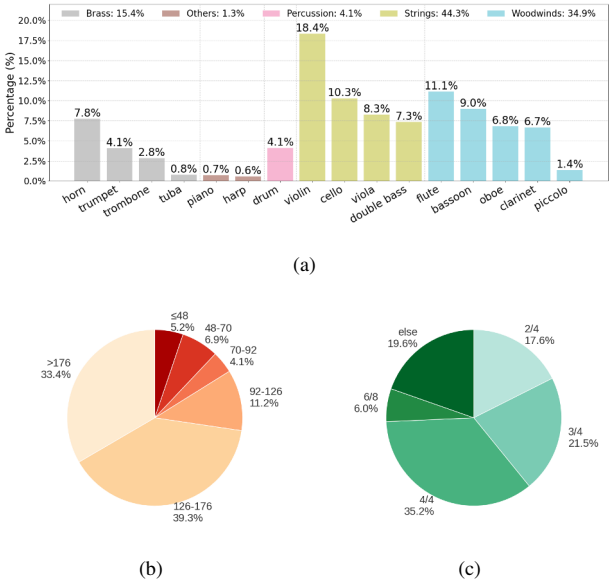


Figure 4: Statics of MSCD. Figure (a) shows the distribution of Instruments. Figure (b) shows the distribution of tempo. Figure (c) shows the distribution of time signatures.

such as the piano and harp, appear infrequently, representing 0.7% and 0.6% respectively.

Different tempos can produce varied musical and emotional effects. In musical notation, tempo markings typically include both BPM (beats per minute) and expressive terms. This analysis focuses solely on the BPM component, categorized into six groups: Largo ($BPM \leq 48$, 5.2%), Adagio ($48 < BPM < 70$, 6.9%), Andantino ($70 < BPM \leq 92$, 4.1%), Allegro ($92 < BPM \leq 126$, 11.2%), Presto ($126 < BPM \leq 176$, 39.3%), and Prestissimo ($BPM > 176$, 33.4%). The distribution of tempo categories is illustrated in Fig. 4(b).

Time signature plays a crucial role in determining the rhythmic structure of a piece by specifying both the number of beats per measure and the beat unit. For example, a 4/4 time signature contains four quarter-note beats per measure. Typically, the first beat is a strong beat, the third is relatively strong, while the second and fourth are weak beats. The conductor's gestures help emphasize these metric patterns by modulating the strength and direction of movement. For a fixed beat pattern, the conductor maintains consistent gestural actions. The distribution of time signatures in the dataset is shown in Fig. 4(c), with the most common being 4/4 (35.2%), 3/4 (21.5%), 2/4 (17.6%), and 6/8 (6.0%). Other signatures, including 3/8 and 5/4, comprise the remaining 19.6%.

Taken together, these statistical insights underscore the comprehensive nature of our dataset. The rich modality coverage—spanning conductor gestures, synchronized scores, and structural musical metadata—positions the dataset as a valuable resource for investigating the nuanced relationships between musical structure and expressive motion. By capturing a wide range of instruments, tempos, and time signatures, the dataset supports both fine-grained analysis and model generalization across diverse musical contexts. This foundation enables future research in areas such as tempo-aware score following, and expressive performance modeling.

5 Experiments

In this section, we present the dataset splits and implementation details, followed by the evaluation metrics used in our experiments. We then report both quantitative and qualitative results, including comparisons with ablation variants and a user study.

5.1 Implementation Details

All experiments are conducted on the MSCD dataset described in Sec. 4. The dataset is divided into training (4.18 hours), validation (0.53 hours), and test (0.53 hours) splits. Each data sample comprises a 10-second video clip aligned with its corresponding symbolic score. Videos are sampled at 25 frames per second, resulting in sequences of 250 frames. Upper-body gesture is represented using 13 keypoints based on the MS-COCO format, producing pose sequences of shape [250, 13, 2]. he aligned music scores are encoded as 1280-dimensional vectors, padded with ¡pad¿ tokens and structured as [1280, 12] matrices.

In the contrastive learning stage, we use a batch size of 32, a learning rate of 3e-4, and train for 800 epochs using the Adam optimizer. In the generative stage, a diffusion process with 1000 steps is applied, using a learning rate of 2e-4, the same batch size, and training for 1000 epochs. A random masking rate of 0.1 is used. The objective weights are set as follows: $\lambda_{ddim} = 10$, $\lambda_{perc} = 0.2$, $\lambda_{geo} = 0.5$, $\lambda_{vel} = 1$ and $\lambda_{elbow} = 1$. All experiments are conducted on two NVIDIA GeForce RTX 3090 GPUs.

5.2 Evaluation Metrics

To assess the quality and diversity of generated gestures, we adopt three evaluation metrics commonly used in gesture synthesis and music-to-motion research:

Mean Squared Error (MSE). MSE measures the average squared distance between predicted and ground-truth joint positions across time. It reflects the model's precision in reconstructing detailed skeletal movements.

Fréchet Gesture Distance (FGD). Adapted from Fréchet Inception Distance, FGD quantifies the distributional similarity between generated and real motions in a learned latent space. It captures global stylistic coherence and gesture realism.

Diversity. This metric evaluates the generative model's ability to produce varied motion outputs. It computes the mean absolute error between 50 randomly sampled generated sequences and their corresponding real motion embeddings, providing a proxy for output variability under limited supervision.

5.3 Experiment Results

This section presents both qualitative and quantitative evaluations of our framework, with ablation studies serving as comparative benchmarks. In the ablation setup, we systematically mask four dimensions of the score encoder: tempo, cycle, intensity, and chord.

5.3.1 Quantitative Evaluation

As shown in Tab. 2, the full-information configuration achieves an Fréchet Gesture Distance (FGD) of 85.396, significantly outperforming all ablated variants. However, although it achieves the best performance in terms of FGD, indicating strong perceptual realism and alignment with natural gesture distributions, the model does not attain the lowest MSE or the highest Diversity score. This discrepancy can be attributed to two key factors.

First, the model prioritizes stylistic coherence and expressive alignment, driven by the contrastive and perceptual loss objectives. As a result, the generated gestures may exhibit slight spatial deviations from the ground-truth trajectories, especially when emphasizing fluidity or dynamic expressiveness. Although the gestures appear perceptually natural, these deviations contribute to higher MSE values, as MSE strictly penalizes point-wise differences without considering semantic or stylistic plausibility.

Second, the relatively lower Diversity score may be explained by the model's reliance on interpretable and structured score features (*e.g.*, tempo, intensity, chord), which tightly constrain the gesture space. Consequently, the model tends to produce consistent and musically grounded gestures for similar inputs, reducing statistical variance across generated samples. While this behavior aligns with

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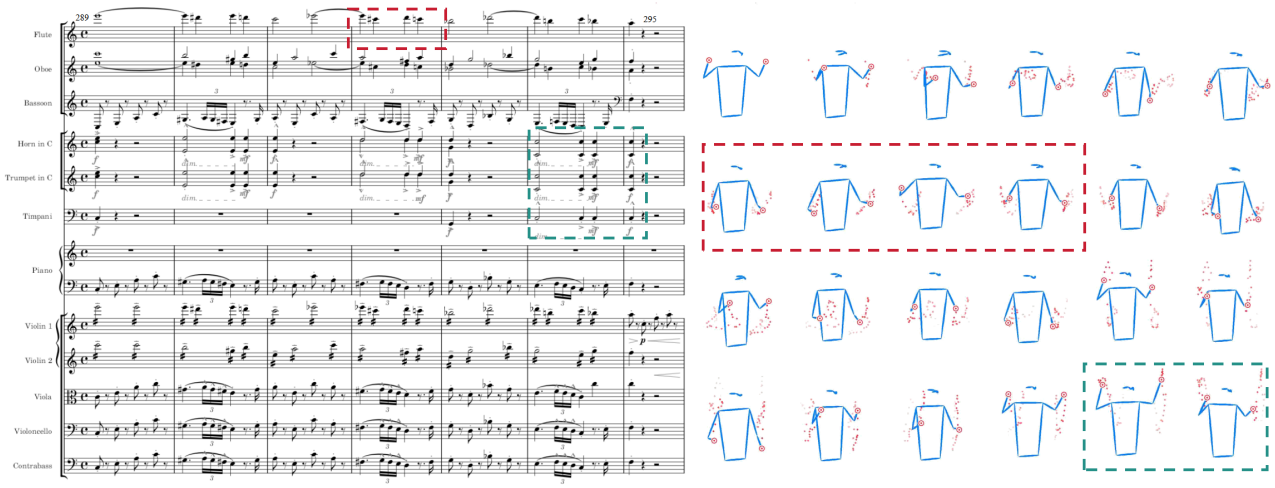


Figure 5: The visualized results of the clip from Mozart's Piano Concerto No. 21 with score. The first bar of the score is the preparation bar, and we compose the gesture sequence by selecting the first frame every 10 frames from the generated result. Where the red border shows that our gestures understand the down beat of a measure and are metronomic in nature, while the tempo remains uniform. The green border shows that our gestures are sensitive to intensity markers and can generate expressive gestures.

real-world conducting conventions, where gestures are expected to reflect the music's structure and emotional contour, it inherently reduces gesture variability, resulting in a trade-off between expressive fidelity and generative diversity.

5.3.2 Qualitative Evaluation

Figure 5 shows the qualitative visualization of generated conducting gestures for *Mozart Piano Concerto No.21*, focusing on bars 290 to 295. Our framework successfully synthesizes complete conducting gesture downbeats, capturing both the temporal continuity and the expressive intent of the musical passage. Notably, the increased gesture amplitude aligned with the score-indicated *forte* markings illustrates the model's capacity to interpret and reflect musical dynamics through physical gesture. The generated trajectories demonstrate smooth transitions and well-articulated directional changes, reinforcing the temporal and emotional alignment between score semantics and gesture.

Furthermore, Fig. 6 qualitatively illustrates the influence of ablating individual score features, providing visual evidence to complement the quantitative results. When chord and cycle features are masked, the resulting gestures become noticeably rigid and repetitive, lacking the subtle temporal modulation characteristic of real conducting. Tempo masking introduces irregular pacing and temporal discontinuities, akin to "speed bumps," which disrupt the natural flow of the gestures. In contrast, removing intensity information reduces the amplitude variation across beats, resulting in flat or under-expressive gestures that fail to mirror musical dynamics. These observations underscore the critical role of each score feature in shaping expressive and musically coherent conducting gestures.

However, we also observe occasional minor artifacts, such as delayed wrist articulation. These imperfections reflect the challenge of tightly synchronizing expressive gestures with symbolic input in real time and suggest future directions for improving temporal modeling and refining the integration of symbolic score semantics.

5.4 User Study

We conducted a user study to evaluate the usability and user satisfaction of our framework by collecting both qualitative and quantitative feedback from participants. We designed an anonymous questionnaire comprising basic questions and evaluation tasks to objectively assess participants' agreement with the generated results and to collect detailed feedback. All procedures complied with standard ethical guidelines, and data were collected only after informed consent was obtained from each participant.

Table 2: Main results on Symphony Conductor Gesture Test-set

methods	FGD	MSE	Diversity
Full	**85.396**	0.020	264.190
Tempo Mask	364.796	0.025	**360.704**
Cycle Mask	299.229	0.038	260.446
Intensity Mask	143.851	**0.014**	223.696
Chord Mask	301.246	0.030	242.941

5.4.1 Experiment Settings

Participants: We primarily recruited participants online, specifically targeting individuals with musical backgrounds to enhance the reliability of our results. In total, we collected 165 responses, with 84 males and 81 females aged 17 to 65. Our participants came from a variety of educational, professional, and social backgrounds. Detailed demographic data are presented in Fig. 7.

Baseline: Since no prior work generates conductor motions driven by the musical score, we adopt the Diffusion-Conductor method [60] as our baseline. Audio signals are extracted from the scores using MuseScore. It is noteworthy that while the audio-driven approach yields outputs at 60 fps, we deliberately refrain from downsampling these results to ensure the reliability of the baseline comparison.

Metrics: During the survey, participants were first given the musical score to review. Afterward, they watched the videos and rated them based on three criteria:

1. **Quality.** Assess whether the overall movement is aesthetically pleasing rather than stiff or awkward.
2. **Semantic consistency.** Evaluate whether the conductor's gestures align with the intended expression of the score.
3. **Readability.** Evaluate whether the gestures are sufficiently clear and structured for orchestra members to follow.

Procedure: For evaluation, we randomly selected three 10-second excerpts from distinct musical pieces (*Haydn Symphony No.8 in G*, *Mozart Piano Concerto No.21*, and *Dvořák Symphony No.9*). To simplify notation, we use Exp_H to denote *Haydn Symphony No. 8 in G*, Exp_M for *Mozart Piano Concerto No. 21*, and Exp_D for

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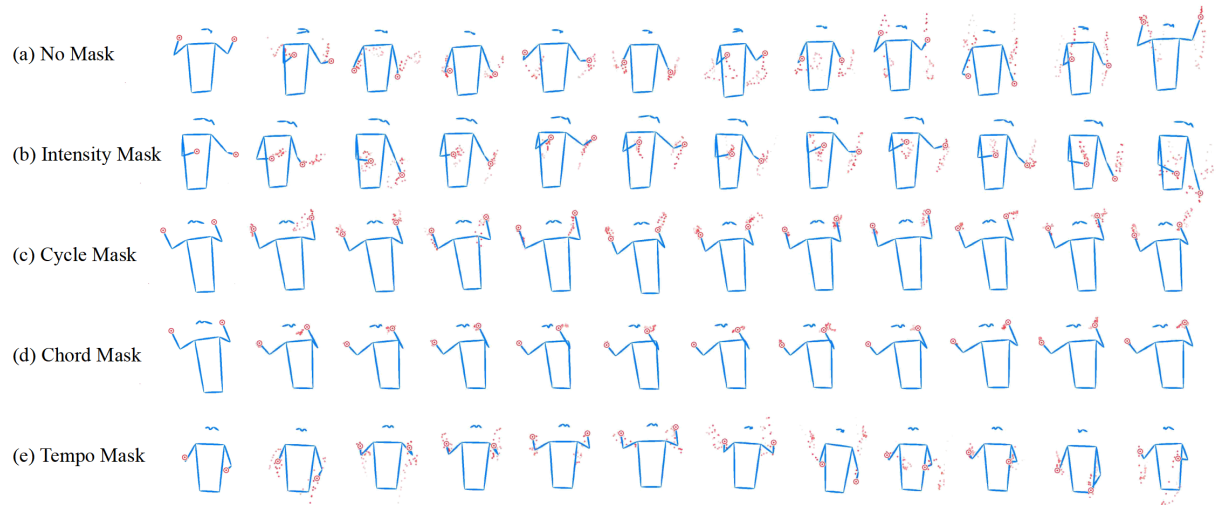


Figure 6: The visual results of the ablation study on a fragment of Mozart's Piano Concerto No. 21 are presented as follows: (a) our method; (b) intensity mask; (c) cycle mask; (d) chord mask; (e) tempo mask. The first frame of each second was chosen to create the gesture sequence. The intensity mask clearly results in a reduced amplitude of the gesture movements. In contrast, the cycle mask and chord mask tend to fixate the gestures, showing little variation in the sequence. The velocity mask, on the other hand, occasionally missed changes, as shown in frames 3-6.

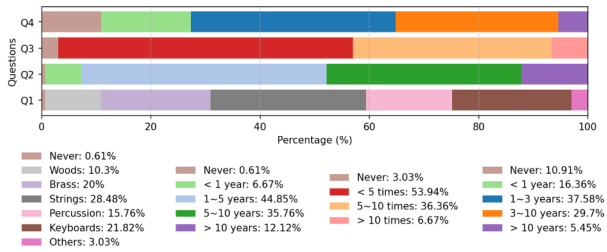


Figure 7: The statistics of general questions: (Q1) Instrument studied; (Q2) Years of study instrument; (Q3) Number of concerts attended; (Q4) Years of symphony orchestra participation.

Dvořák Symphony No. 9. Each excerpt underwent six comparative experimental conditions (audio-driven, our proposed method, and four ablation variants), with presentation order randomized per participant. Each participant rated the 18 video clips (3 pieces × 6 conditions) across three evaluation metrics on a scale of 0 to 100. Participants viewed 2D skeletal animations of conductor motions representative format shown in Fig. 6.

5.4.2 Statistical Analysis

After removing invalid responses, 163 valid assessments were retained for analysis. We conducted repeated measures ANOVA for audio-driven and our model, as well as the comparison between the model and the four ablation experiments. We formally propose the following null hypotheses (H_0) and research questions:

- H_{01}: There are no significant differences in the mean scores between the baseline model and our model in the target metrics.
- H_{02}: There are no significant differences in the mean scores between the ablated models and our model in the target metrics.

Derived from these, research questions guide our investigation:

- RQ1: Does performance significantly differ between the baseline and our model?
- RQ2: Do all four components significantly enhance the effectiveness of our model?

Quality Evaluation. This evaluation aims to assess whether the generated results meet the aesthetic expectations. For Exp_H, Exp_M, and Exp_D, our model yielded mean scores of 65.98, 64.68, and 65.65, respectively, compared to 79.57, 77.10, and 78.11 for the audio-driven method in each experiment. The difference was not statistically significant, as revealed in an analysis of variance (Exp_H: $F_{1,162} = 0.122$, ns; Exp_D: $F_{1,162} = 0.023$, ns; Exp_M: $F_{1,162} = 1.566$, $p > .05$). The comparison between our model and the four ablation experiments showed that the main effect of model type was significant ($p < .05$). Due to within-subject comparisons, Bonferroni correction was applied ($\alpha = 0.005$) [53], and the effects of all comparison groups reached significant levels ($p < 0.005$).

Four ablation experiments confirmed the effectiveness of each component of our model. Notably, it performs better in strong dynamic contrasts, such as sudden shifts from "sforzando" to "piano," allowing for compelling amplitude gradients through explicit intensity encoding. However, while statistical tests show no significant difference from the baseline model (with $p > .05$), some indicators trend lower for our method. This can be attributed to two key trade-offs: our use of a 25 fps frame rate, compared to the baseline's 60 fps, limits the continuity of fine movements and leads to more mechanical interpolation in fast-paced passages. Additionally, the implicit modeling strategy of the score encoder improves dynamic expression but falls short in capturing the geometric regularity of standard beat patterns compared to the explicit audio-driven method. These factors contribute to the observed differences in performance.

Semantic Consistency. This evaluation aims to assess whether the generated results meet the aesthetic expectations. For Exp_H, Exp_M, and Exp_D, our model yielded mean scores of 64.89, 64.39, and 64.19, respectively, compared to 78.90, 74.61, and 78.16 for the audio-driven method in each experiment. The difference was not statistically significant, as revealed in an analysis of variance(Exp_H: $F_{1,162} = 0.511$, ns; Exp_D: $F_{1,162} = 0.560$, ns; Exp_M: $F_{1,162} = 0.851$, ns). The comparison between our model and the four ablation experiments showed that the main effect of model type was significant ($p < .05$). Due to within-subject comparisons, Bonferroni correction was applied (the corrected $\alpha = 0.005$), and the effects of all comparison groups reached significant levels ($p < 0.005$).

While audio signals inherently encode continuous temporal dynamics, the symbolic nature of score input requires us to reconstruct such continuity through explicit strategies. Our hierarchical encod-

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ing strategy effectively enhances local semantic fidelity, particularly in the more precise capture of discrete symbolic cues such as intensity and downbeats. Specifically, through the explicit mapping strategy, the model demonstrates unique advantages in handling such discrete features. It should be noted, however, that in terms of global temporal consistency, the model's coupled modeling of time-sensitive features, such as rhythmic cycles, still has room for optimization, which may lead to slight room for improvement in the semantic coherence of gesture expressions under complex timing. Overall, this hierarchical design has formed a unique technical approach to balancing symbolic feature parsing and dynamic continuity reconstruction, mainly providing a valuable solution for the digital expression of music in the precise processing of discrete symbols.

Readability Evaluation. This evaluation assesses the clarity and interpretability of generated gestures for professional orchestra members. For Exp_H, Exp_M, and Exp_D, our model yielded mean scores of 67.93, 66.93, and 68.08, respectively, compared to 82.04, 78.98, and 81.12 for the audio-driven method in each experiment. The difference was not statistically significant, as revealed in an analysis of variance(Exp_H: $F_{1,162} = 0.829$, ns; Exp_D: $F_{1,162} = 0.463$, ns; Exp_M: $F_{1,162} = 0.654$, ns). The comparison between our model and the four ablation experiments showed that the main effect of model type was significant ($p < .05$). Due to within-subject comparisons, Bonferroni correction was applied ($\alpha = 0.005$), and the effects of all comparison groups reached significant levels ($p < 0.005$).

The base-work method improves readability by synchronizing closely with the audio signal, while our framework has difficulty representing beat structures, leading to ambiguities in gesture boundaries. Nonetheless, our approach excels in multi-instrument contexts. By utilizing symbolic score information, we can provide individualized cues for different instrument sections, which is a capability that audio-driven methods lack. This mapping allows for hierarchical signaling, crucial for coordinating nuanced orchestral passages.

Conclusion. Based on the analysis, we conclude the following: H01 is valid, indicating no statistically significant difference in mean scores between our model and the baseline model. H02 is rejected, showing a significant difference in mean scores between our proposed model and the four ablated models. For RQ1, we find no notable distinction. For RQ2, we conclude that each component significantly contributes to our model's improvement.

5.5 User Feedback

We also collected open-ended feedback from the participants, which provided several insightful comments. Many amateur participants expressed a strong interest in the virtual conducting experience. A significant number reported having participated in school orchestras and noted that it is often challenging to recapture the immersive experience of conducting after leaving the school environment. They feel that virtual conducting helps bridge this gap. Additionally, several participants suggested that an online version of the system could greatly expand its practical applications and reach a wider audience.

Discussions with instrumentalists in symphony orchestras highlighted technical critiques of our conducting methods. While they acknowledged the strengths of the concept of score-driven conducting, they found it less effective than the Diffusion-Conductor method in terms of beat precision and movement smoothness. However, they noted that our approach is more adaptable to instrumentation changes and offers a wider variety of conducting gestures, which is also proven in our quantitative findings.

Furthermore, professionals highlighted a critical limitation shared by all current systems: the absence of real-time adaptability. In actual performance settings, conductors often adjust their gestures in response to ensemble dynamics and unexpected musical events, a capacity not yet addressed by existing methods.

6 Discussion

6.1 Future Work

While our work facilitates the generation of conductor gestures from musical scores, there are several important research directions that require further exploration:

High-Fidelity Symphony Conductor Gesture Dataset. The main goal is to create a high-quality dataset. Although our MSCD effectively analyzes scores and generates gestures, frame-rate limitations hinder performance. The lack of publicly available web data also makes it hard to gather diverse musical samples. To overcome these issues, we will use motion capture techniques to improve our dataset and prepare resources for multi-model learning.

Refined Conductor Gesture Modeling. Enhanced datasets warrant the exploration of more adaptable and interactive gesture generation models. Current solutions overlook critical nuances, such as facial expressions [30], eye contact [45], which are important for effective communication in conducting. Virtual conductors should incorporate these subtle gestural dimensions.

Real-Time Responsiveness. A crucial goal is to enable conductors to adapt in real-time to feedback from live ensembles. A complete conducting system should establish a closed-loop chain: score $\rightarrow$ conductor gestures $\rightarrow$ performer execution $\rightarrow$ gesture refinement through auditory feedback. Current systems lack this capability, so implementing such feedback loops is essential for the practical deployment of conducting gesture systems.

6.2 Application

We developed an interactive demonstration of a virtual conductor system in virtual reality (VR), illustrated in Fig. 1. Built on HTC Vive devices using Unreal Engine (UE), the system includes concert hall scene design, user interface interaction, and a sheet music reading module. To overcome implementation challenges, we proposed several solutions: (1) Native annotation tools for real-time MusicXML conversion via controllers or voice commands to enhance sheet music editing. (2) Geometric-acoustic algorithms for authentic orchestral spatial audio rendering based on concert hall parameters. (3) Score-audio multimodal learning to enable performance-adaptive gestures for real-time conducting responsiveness. Additionally, we will focus on system scalability, including instrument-specific gesture control, multiplayer synchronization, and educational modules that connect gestures with scores for improved accessibility.

7 Conclusion

This paper introduces a novel score-based method for generating virtual conductor gestures in a virtual reality VR system, utilizing a two-stage architecture. The first stage involves a contrastive pre-training phase to learn meaningful score representations, followed by a generative phase using a Denoising Diffusion Implicit Model (DDIM) to synthesize conductor gestures. We also present the Music Score-Conductor Dataset MSCD, which provides aligned score-gesture pairs to address a major gap in the field. Through quantitative evaluations and user studies with amateur and professional musicians, we validate our approach's effectiveness and limitations. Although our method currently falls short of audio-based models in motion fluidity, it excels in instrument-specific cueing and intensity-aware variability. Our research establishes a foundation for a new class of virtual conducting systems that could enhance music education and expand the expressive capabilities of virtual orchestras.

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